1.- ¿Cuantos gramos de soluto se necesitan para preparar las siguientes disoluciones?

a) 2l de solución de cloruro de sodio 10% $\frac{m}{V}$

b) 500g de disolución de $Hg(OH)_2$ al 2% $\frac{m}{m}$

c) 100ml de disolución de Ioduro de potasio al 12% m/m con d = 1,118 g/ml

a) $10\% \frac{m}{V} = \frac{10 g\ NaCl}{100 ml\ sl\acute{u}n}$

$100 ml\ slc \longrightarrow 10 g\ NaCl$

$2000 ml\ slc \longrightarrow x = \frac{2000 \cancel{ml\ slc} \times 10 g\ NaCl}{100 \cancel{ml\ slc}}$

$\boxed{x = 200 g\ NaCl}$

Otra forma

$\frac{10 g\ NaCl}{100 \cancel{ml\ slc}} \times 2000 \cancel{ml\ slc} = \boxed{200 g\ NaCl}$

b) $500 g\ Hg(OH)_2$ al $2\% \frac{m}{m}$

$100 g\ slc \longrightarrow 2 g\ Hg(OH)_2$

$500 g\ slc \longrightarrow x = \frac{500 g\ \cancel{slc} \times 2 g\ Hg(OH)_2}{100 g\ \cancel{slc}}$

$\boxed{x = 10 g\ Hg(OH)_2}$

c)

c) $12\% \frac{m}{m}$ $d = 1,18 \frac{g}{ml}$

g IK? 100 ml slc

$\% \frac{m}{m} \longrightarrow \% \frac{m}{V}$

$\frac{g\,sto}{100\,g\,slc} \xrightarrow{\times d\,slc} \frac{g\,sto}{100\,ml\,slc}$

$\frac{g\,sto}{100\,\cancel{g}\,slc} \times \frac{\cancel{g}\,sol}{ml\,sol} = \frac{g\,sto}{100\,ml\,slc}$

$12\% \frac{m}{m} \times 1,18 \frac{g}{ml}\,slc = 14,16 \frac{g\,sto}{100\,ml\,slc}$

Para preparar 100 ml slc $\xrightarrow{\text{se necesitan}}$ $\boxed{14,16\,g\,IK}$

2) Una disolución contiene 2g de soluto en 48g de agua. Expresar su concentración en % m/m y ppm

$$\% \frac{m}{m} = \frac{g\ sto}{100g\ slc}$$

$$ppm = \frac{mg\ sto}{kg\ slc}$$

$$\left.\begin{array}{l} 2g\ sto \\ 48g\ H_2O \end{array}\right\} m\ slc = 2g\ sto + 48g\ H_2O = 50g\ slc$$

$2g\ sto \rightarrow 50g\ slc$

$4g\ sto = x \leftarrow 100g\ slc$

$$\Rightarrow \frac{4g\ sto}{100g\ slc} = \boxed{4\% \frac{m}{m}}$$

$2g\ sto = 2000mg\ sto$

$2000mg\ sto \rightarrow 0{,}05kg\ slc$

$40000\ mg = x \leftarrow 1kg\ slc$

$\Rightarrow \boxed{40\,000\ ppm}$

Problemo Nº 3

24,4 g Na(OH)

100 mL scion

$\rho_{scion} = 1{,}22$ g/mL

$\%m/V = \frac{g\ sto}{100\,mL\ scion} = \frac{24{,}4\ g}{100\ mL\ scion}$

$\%m/V = \boxed{24{,}4\ (\%m/V)}$

$g/L = \frac{g\ sto}{L\ scion} = \frac{24{,}4\ g\ Na(OH)}{0{,}1\ L} \Rightarrow \boxed{244\ (g/L)}$

%m/m ? → se relaciona con la densidad

$\frac{\%m/m}{\%m/V} = \frac{\frac{g\ sto}{100\ mL\ scion}}{\frac{g\ sto}{100\ g\ scion}} = \frac{\cancel{g\ sto} \cdot 100\ g\ scion}{\cancel{g\ sto} \cdot 100\ mL\ scion} = \frac{g\ scion}{mL\ scion}$

$\rho = \frac{m}{V} = \frac{g}{mL}$!!! $\Rightarrow \boxed{\frac{\%m/m}{\%m/V} = \rho_{scion}}$

$\Rightarrow \%m/m = \%\frac{m}{V} \cdot \rho_{scion} = 24{,}4 \cdot 1{,}22 \Rightarrow \boxed{29{,}77\ \%m/m}$

Problema Nº4

a) 250 mL H_2SO_4 al 2 M →

$M = \frac{n_{soluto}}{L\ scion} \Rightarrow n_{sto} = M \cdot L_{scion}$

$n_{sto} = 2 \cdot 0{,}250$

$n = 0{,}5$ mol

1 mol H_2SO_4 → 82 g

0,5 mol H_2SO_4 → $\boxed{X = 41\ g\ H_2SO_4}$

b)

b) 2 kg scion $MgCl_2$ al 0,1 m

$$m = \frac{n_{sto}}{kg\ ste}$$

$$m_{sción} = m_{sto} + m_{ste} \Rightarrow m_{ste} = m_{sción} - m_{sto}$$

$$m \cdot kg_{ste} = n_{sto} \Rightarrow \boxed{m \cdot (kg_{scion} - kg_{sto}) = n_{sto}} \; (*)$$

Pero $MM_{MgCl_2} = \frac{m_{sto}}{n_{sto}} \rightarrow n_{sto} = \frac{m_{sto}}{MM_{MgCl_2}}$ → reemplazo en (*) y paso todo a kg

$$m \cdot kg_{scion} - m\, kg_{sto} = \frac{kg_{sto}}{MM_{sto}}$$

$$MM_{sto} \cdot m\, kg_{scion} - MM_{sto} \cdot m\, kg_{sto} = kg_{sto}$$

$$\left(\frac{MM_{sto} \cdot m \cdot kg_{scion}}{1 + MM_{sto} \cdot m}\right) = kg\ sto$$

$$kg_{sto} = \frac{95 \times 10^{-3} \left(\frac{kg}{mol}\right) \cdot 0{,}1 \left(\frac{mol}{kg}\right) \cdot 2\,(kg)}{\left(1 + 95 \times 10^{-3} \frac{kg}{mol} \cdot 0{,}1 \frac{mol}{kg}\right)} = 0{,}0188\ kg$$

$$\boxed{18{,}82\ g\ MgCl_2}$$

Verifico $m = \frac{n}{kg_{ste}} = \frac{18{,}82/95}{2 - 18{,}82 \cdot 10^{-3}} = 0{,}1$

Problema N° 5

* $m = ?$
* $X = ?$
* Scion Na_2CO_3 al 20% m/V

$\rho = 1{,}125$ g/mol

1°) Supongamos 100 mL scion !!!

si está al 20% m/V; en 100mL hay

$\boxed{20\,g\ Na_2CO_3}$

Si 1mol Na_2CO_3 $\xrightarrow{pesa}$ 106 g

$\boxed{0{,}189\ mol = X}$ ← 20 g

$\Rightarrow M = \frac{n_{sto}}{L_{scion}} = \frac{0{,}189\ mol}{0{,}1\ L}$

$\boxed{M = 1{,}88\ M}$

2°) Además $\Rightarrow$ $m = \frac{n}{kg_{ste}}$ en kilos

$\Rightarrow \rho_{scion} = \frac{m_{scion}}{V_{scion}}$

$m_{scion} = V_{scion} \cdot \rho_{scion}$

$m_{scion} = 100\ mL \cdot 1{,}125\ \frac{g}{mL} = \boxed{112{,}5\ g}$

Pero $m_{ste} = m_{scion} - m_{sto}$

$m_{ste} = 112{,}5\ g - 20\ g \Rightarrow \boxed{92{,}5\ g\ H_2O}$

$\Rightarrow m = \frac{0{,}189\ mol}{92{,}5 \times 10^{-3}\ kg} \Rightarrow \boxed{m = 2{,}04\ m}$

finalmente

$X = \frac{n_{sto}}{n_{ste}} = \frac{0{,}189\ mol\ Na_2CO_3}{5{,}14\ mol} \Rightarrow \boxed{X = 0{,}037}$

1mol H_2O → 18 g

$= X$ ← 92,5 g

5,14 mol

Escaneado con CamSca

Problema N° 6

H_2SO_4 al 65 %m/m
$\rho = 1{,}126 \frac{g}{mL}$

Supongo 100 gr Sción

$\Rightarrow \%m/m = \frac{gr\ sto}{100\,g\ scion} \Rightarrow \boxed{Hay\ 65\,g\ sto}$

además si $\rho_{scion} = \frac{m}{V} \Rightarrow V_{scion} = \frac{100\ gr\ scion}{1{,}126 \frac{gr}{mL}} = \boxed{79{,}36\ mL\ scion}$

$\Rightarrow n_{sto}$? 1mol H_2SO_4 — 98 g
$\boxed{0{,}66\,mol = X} \leftarrow 65\,g \Rightarrow$

$M = \frac{n_{sto}}{L_{scion}} = \frac{0{,}66\,mol}{79{,}36 \cdot 10^{-3}\,L} \Rightarrow \boxed{M = 8{,}35\ Molar}$

$m = \frac{n_{sto}}{Kg\ ste}$

$m_{ste} = m_{scion} - m_{sto} = 100\,g - 65\,g = 35\,g\,ste$

$m = \frac{0{,}66\,mol}{35 \cdot 10^{-3}\,Kg} = \boxed{18{,}85\ m}$

Problema N°7

$M_{sto} = 50\,g$ KCl

$m_{ste} = 850\,g$

$\rho_{scion} = 1{,}195\,g/mL$

$M_{scion} = 850\,g + 50\,g = \underline{900\,g\ scion}$

$V_{scion} = \dfrac{m}{\rho} = \dfrac{900\,g}{1{,}195} = \underline{753{,}14\ mL\ scion}$

$\%\,m/V = \dfrac{50\,g}{7{,}5314 \times 100\,mL\ scion} \Rightarrow \boxed{6{,}64\ \%\,m/V}$

$\%\,m/m = \dfrac{50\ g\,sto}{9 \times 100\,g\,scion} \Rightarrow \boxed{5{,}55\ \%\,m/m}$

n_{sto}?

1 mol KCl $\rightarrow$ 74,5 g

$\boxed{0{,}67\ mol = X} \leftarrow$ 50 g

$M = \dfrac{0{,}67\ mol}{0{,}753\ L\,scion}$

$\boxed{M = 0{,}89\ Molar}$

M_{ste}?

1 mol H_2O $\rightarrow$ 18 g

47,22 mol = X $\leftarrow$ 850 g

$m = \dfrac{0{,}67\ mol}{0{,}850\ kg\ scin} = \boxed{0{,}78\,m}$

$X = \dfrac{0{,}67\ mol\ KCl}{47{,}22\ mol\ H_2O} = \boxed{X = 0{,}014}$